

Happy 70th Birthday CERN!



CERN flags are flying on the Mont Blanc Bridge in Geneva from 1 to 6 October to celebrate the 70th birthday of the Organization. There is also a [temporary CERN exhibition](#) at the [Rotonde du Mont-Blanc](#) from 1 to 31 October. (Image: CERN)

A special high-level ceremony on [1 October](#) crowned a year of celebrations for CERN's 70th anniversary, which are [continuing](#) across Europe and beyond.

The ceremony, held at CERN Science Gateway, was attended by 38 national delegations and included speeches by Heads of State and Government from Bulgaria, Italy, Latvia, Serbia, Slovakia and Switzerland, Her Royal Highness the Princess Astrid of Belgium, and the President of the European Commission. In addition to the national delegations, many scientific, political and economic leaders were also present.

A series of short films highlighted significant milestones in CERN's history, and the event featured a performance from the *Orchestre de la Suisse Romande*.



CERN Director-General, Fabiola Gianotti, gave the opening speech at CERN's 70th anniversary ceremony, which took place in CERN Science Gateway (watch the [recording](#) from timecode 13:55 to see her speech in full). (Image: CERN)

A Word from Fabiola Gianotti

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Robert Aymar (1936 – 2024)

“Today, CERN is a truly global laboratory, with 24 Member States (twice as many as in 1954), ten Associate Member States, four Observers, many partners from all over the world and a community of more than 17 000 people across more than 110 nationalities, an incredible melting pot of languages, cultures and traditions. For those of us who work here – or have worked here – this rich and enriching diversity has become a normal part of our everyday life. But in a world where conflicts between countries, religions and cultures sadly persist, this is a truly precious gift that cannot be taken for granted.”

- Fabiola Gianotti, CERN Director-General



Her Royal Highness Princess Astrid of Belgium and His Royal Highness Prince Lorenz of Belgium (centre) visited the underground CMS experimental cavern before the ceremony on 1 October. They are joined here by Benoît Delille, Head of the Occupational Health & Safety and Environmental Protection (HSE) unit (left) and Gautier Hamel de Monchenault, Spokesperson for the CMS experiment (right). (Image: CERN)



Family picture taken at CERN's 70th anniversary ceremony (Image: CERN)

This high-level group of distinguished guests demonstrated the strong support for CERN's mission and its ambitions.

“There was so much positive feedback from all the delegations and guests and the support that was expressed by them for the future of CERN was extremely moving”, says Charlotte Lindberg Warakaulle, CERN Director of International Relations. “This was an outstanding event in a truly unique year of celebration. What a great basis for a brilliant future!”

The [year-long anniversary programme](#) continues to grow and has already seen more than 100 events organised in 29 countries as well as at CERN. “We are truly grateful to everyone who has contributed throughout the year,” said Luciano Musa and Melissa Gaillard, Coordinator and Deputy Coordinator of the CERN70 events. “The feedback that we continue to receive from members of the CERN community is also amazing. Thank you to everyone who was involved.”



President of the European Commission, Ursula von der Leyen, was one of [the high-level speakers](#) at the ceremony. She gave an impassioned speech in support of CERN and its future (watch the [recording](#) from timecode 37:25 to see her speech in full). (Image: CERN)

Celebrations also took place on social media, including a video of the CERN community wishing the Laboratory a [multilingual Happy Birthday](#), and even birthday messages sent in by [celebrities from around the world](#).

The full recording of the ceremony is available on CERN's [YouTube page](#).

Photo highlights from the ceremony are available [here](#).

Internal Communication

A Word from Fabiola Gianotti

From September Council to seventieth celebrations

Last week, during an intense and interesting Council session, important discussions relating to the future of CERN were held

Last week, during an intense and interesting Council session, important discussions relating to the future of CERN were held. Some aspects of the process leading to the [update of European Strategy for Particle Physics](#), namely the establishment of the Physics Preparatory Group (PPG) and the selection of the venue for the Open Symposium (Venice, Italy), were finalised, and the Council elected Professor [Costas Fountas](#) (University of Ioannina) as its 25th President, taking over from Professor Eliezer Rabinovici on 1 January 2025.

Council delegates congratulated the CERN community for the outstanding performance of the accelerator complex, the experiments and computing. The LHC reached its 2024 luminosity-production [target](#) (110 fb^{-1} for ATLAS and CMS) some 2.5 weeks ahead of schedule.

In addition, the Council and its subordinate bodies were informed of a [revised schedule for Long Shutdown 3 \(LS3\)](#). The LHC will now enter LS3 in July 2026, seven and a half months later than planned, with the start of the High-Luminosity LHC (Run 4) beginning in June 2030. The injectors will run until the end of August 2026. LS3 is a critical activity in ensuring the continued full exploitation of CERN's physics potential in the coming decades.

CERN's brilliant past achievements and bright future were centre stage during a [special high-level event](#) on Tuesday this week, when 38 national delegations, including eight heads of state or government and 13 ministers, along with many scientific, political and economic leaders, came to CERN to celebrate our unique institution and its 70 years of science, innovation and collaboration. Distinguished guests demonstrated strong support for CERN's mission and future ambition (in particular, the [Future Circular Collider](#)), acknowledging its role not just as the world's foremost high-energy physics laboratory but also as a centre for the development of advanced technologies, a platform for training and education and an exceptional example of collaboration across borders.

Since January, [CERN's 70th anniversary programme](#) has engaged thousands of people around the world with the wonders and applications of particle physics. On behalf of the Directorate, I wish to thank everyone who has contributed to this fantastic success and the whole community for showing CERN at its very best.

Fabiola Gianotti
Director-General of CERN

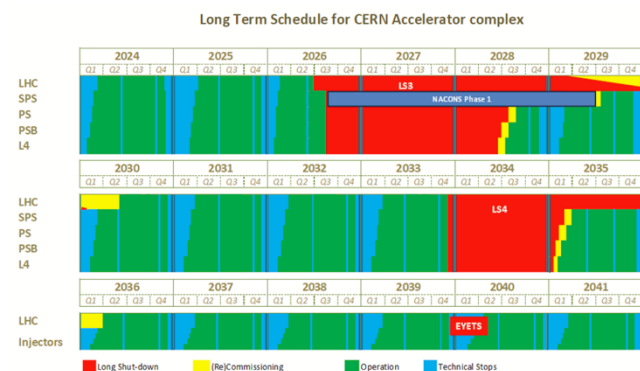
A Word from Mike Lamont

Updated schedule for CERN's accelerators

The timing and duration of the third long shutdown (LS3) of the Large Hadron Collider and injector complex has changed, here's why

In the September CERN Council Session, we presented a revised schedule for Long Shutdown 3 (LS3) of the Large Hadron Collider (LHC) and its injector complex. This new schedule includes a delayed start and an increased duration for LS3. LS3 for the LHC is now scheduled to begin at the start of July 2026, seven and a half months later than planned. The overall length of the shutdown will increase by around four months. Combined,

these measures will shift the start of the High-Luminosity LHC (HL-LHC) (Run 4) by approximately one year, to June 2030. The injectors will continue to operate throughout July and August 2026 and start their extensive programme of work in September 2026. The injectors will gradually restart operations in 2028; a more detailed plan will be established in the coming weeks.



(Image: Jean-Philippe Tock)

Complex challenges and collaborative solutions

The delayed start of LS3 is primarily due to significant challenges encountered during the Phase II upgrades of the ATLAS and CMS experiments, which have led to the erosion of contingency time and introduced considerable schedule risks. The challenges faced by the experiment teams include COVID-19 and the impact of the Russian invasion of Ukraine. For ATLAS, the new Inner Tracker (ITk) is on the critical path and, for CMS, the tracker and HGCal projects are currently operating with low schedule contingency.

Preparing for high luminosity

On the machine side, a key activity during LS3 is the drilling of 28 vertical cores to link the new HL-LHC technical galleries to the LHC tunnel. Although initially expected to take six months, this timeframe was reduced to two months in 2021 to optimise the schedule. However, challenges encountered during the tendering process and in subsequent consultations with specialists necessitated a return to the original six-month timeline for core excavation. The decision to shift the start of the HL-LHC by approximately one year and increase the length of the shutdown reflects a consensus supported by our scientific committees. The decision was reached through careful dialogue that began in the middle of this year, involving machine teams, experiments and the CERN Directorate.

Major upgrades: paving the way for future exploitation

LS3 represents a pivotal phase in enhancing CERN's capabilities, involving substantial improvements aimed at transforming the LHC into

the HL-LHC, as well as major upgrades to both the ATLAS and CMS experiments. These upgrades are designed to significantly enhance the luminosity and data collection capabilities of the experiments. They will enable a diverse and extensive physics programme that will ensure that CERN remains at the forefront of high-energy physics into the early 2040s.

In addition to high-luminosity enhancements, LS3 will involve a huge programme of work across the accelerator complex and encompass numerous critical projects, including:

- **Phase 1 of the North Area consolidation project** to enhance our long-term fixed target research capabilities. Work will start on transforming **ECN3 into a high intensity fixed target facility**.
- **CNGS target dismantling** to pave the way for enhanced wakefield acceleration research at AWAKE.
- **ISOLDE improvements**, including replacing ageing beam dumps, to enable long-term exploitation and boost the facility's nuclear studies potential.
- Extensive maintenance and consolidation across all machines and facilities, including renovation of personnel safety systems and technical galleries, to ensure operational safety, longevity and availability.

All these activities are essential to ensuring the medium-term future of the Laboratory and allowing full exploitation of its remarkable potential in the coming decades. The LS3 Readiness Review, which took place from 11 to 13 September, underscored the scale of this endeavour and our advanced state of preparedness, leveraging considerable experience from previous shutdowns, committed teams and stakeholder engagement to ensure the success of all these extensive upgrades and modifications. As we reflect on the exceptional performance of our entire accelerator complex and its technical infrastructure – highlighted by the 110 fb^{-1} collected by ATLAS and CMS so far this year – it is important to acknowledge that the extensive work performed during long shutdowns and technical stops is fundamental to our continued success.

Mike Lamont

Director for Accelerators and Technology

Accelerator Report: Target reached, but compressor off

On 23 September, the LHC reached its 2024 target for integrated luminosity. And then a cryogenic station broke down...

In the [last report](#), I highlighted that the LHC had successfully delivered 100 fb^{-1} of integrated luminosity to ATLAS and CMS by 13 September 2024. Only ten days later, on 23 September, an additional 10 fb^{-1} was integrated, reaching the 110 fb^{-1} target set for 2024.

Shortly after achieving this milestone, however, one of the LHC's static VAR compensators (SVC, see box) switched off unexpectedly. Fortunately, this didn't affect the beams already circulating in the LHC.

SVCs play a crucial role in stabilising the voltage of the electrical grid, particularly when large pulsed systems – like the magnets located in the transfer line between the SPS and the LHC – are used to inject new beams into the LHC. To inject new beams, it was necessary to turn this SVC back on. Experts were called in and they followed the established procedure, which had worked reliably in the past. This process involves reducing the electrical load by temporarily switching off some equipment as a precaution.

Despite these steps, an electrical perturbation occurred, leading to the shutdown of one of the two cryogenic compressor stations at LHC Point 8, which could not be restarted due to the damage it had sustained. Both of these stations supply liquid helium to cool down an arc of the LHC. With one of the two compressors out of action, the cooling capacity was severely reduced.

As a result, the number of bunches that could circulate in the LHC should have been significantly reduced, from the usual 2352 bunches per beam to about 1500 bunches per beam. This reduction would have affected the luminosity production significantly.



Damaged wheel on the Cold Compressor #2

The damaged compressor. (Image: CERN/TE-CRG)

Fortunately, the cryogenics experts evaluated the situation and, miraculously, managed to exchange the broken cold compressor using a procedure that had never been used in a real-life situation before and without completely warming up the compressor station, saving precious time. In the afternoon of 25 September, the LHC's cryogenic system was fully restored, allowing beams to be injected again.

However, a series of unrelated small technical issues followed. This meant that efficient luminosity production could not be resumed until the scheduled stop, on the evening of 26 September, when the fourth Machine Development (MD) block began.



A static VAR compensator (not the one that switched off). (Image: CERN)

The MD period concluded successfully at midnight on 30 September, just in time to prepare for the VIP visits on the morning of 1 October, organised as part of CERN's 70th anniversary.

Luminosity production was resumed later that day and will continue up until the end of the proton physics run, scheduled for 17 October. After that, the lead-ion physics period will start.

Rende Steerenberg

Static	VAR	compensator:
A static VAR compensator (SVC) is a system used to regulate and stabilise the voltage in CERN's high-voltage electrical network. It dynamically manages the flow of reactive power, either injecting or absorbing it from the grid, as needed. This real-time control is crucial for maintaining voltage stability, particularly under varying load conditions, such as those caused by the cycling of the CERN accelerators.		

CERN70: Switching on the Large Hadron Collider

Part 18 of the [CERN70](#) feature series. Lyn Evans was Project Leader for the Large Hadron Collider



Some of the first magnets installed in the LHC tunnel in 2005. (Image: [CERN](#))

The beginning of the [Large Hadron Collider \(LHC\)](#) dates back to the early 1980s. CERN's major accelerator, the [Large Electron Positron Collider \(LEP\)](#), was still in its [study phase](#). But scientists were already considering using its 27 km tunnel for a proton collider.

Hadron colliders were indeed extremely promising. CERN had developed revolutionary techniques with the [ISR](#), the [world's first proton-proton collider](#) in the 1970s and was preparing a [proton-antiproton collider](#) that was to offer a [major discovery](#) in 1983. On the other side of the Atlantic, a gigantic superconducting supercollider (SSC) project was taking shape.

The LHC was officially launched at a symposium in Lausanne in 1984. The consensus was quickly reached on a high-luminosity machine – with a high number of collisions – to compensate for the lower energy compared with the American mega project. This constraint meant using two very intense proton beams and therefore two rings of magnets. In addition, achieving the required magnetic field meant using innovative [superconducting magnets](#).

These requirements led to the adoption of an ambitious “two-in-one” magnet design: two superconducting coils surrounding the two beam tubes within a single mechanical structure. The superconducting material chosen was niobium-titanium, enabling a magnetic field higher than 8 Teslas, and therefore a collision energy higher than 10 TeV for the protons. The choice of superconductivity meant that the magnets had to be cooled to -271°C using superfluid helium.

After several years of development, prototype magnets were operating successfully in 1994. At the same time, ideas for experiments were taking shape. At a meeting in Evian in 1992, scientific teams submitted their first proposals, laying the foundations for future major international collaborations and their detectors.

But the scientific community and CERN's Member States still had to be convinced. [Carlo Rubbia](#), Director-General of CERN between 1989 and 1994, was one of the great promoters of the LHC, winning the support of the particle physics community in Europe. Consensus was harder to achieve among CERN's Member States. At the end of 1994, the CERN Council approved the LHC, but in two stages.

This option was largely impractical, but a major event the previous year was to unblock the situation. The US Congress had halted the SSC project, leaving the LHC alone to conquer the high-energy frontier. To raise additional funds, the CERN management, led by the new Director-General [Chris Llewellyn Smith](#), conducted



The LHC is a global machine, with contributions from Canada, India, Japan, Russia and the United States. These focusing magnets, for example, were manufactured by the American laboratory Fermilab. (Image: Peter Ginter/CERN)



Five CERN Director-Generals helped the LHC to go from idea to reality, and were all present to witness the first beam on 10 September 2008. Left to right: [Robert Aymar](#), [Luciano Maiani](#), [Christopher Llewellyn Smith](#), [Carlo Rubbia](#) and [Herwig Schopper](#). (Image: CERN)

an intense diplomatic campaign with non-Member States of CERN, including those involved in the SSC project. Canada, India, Russia and Japan signed participation agreements. At the end of 1996, the Council approved the construction of the LHC in one stage. In 1997, the United States joined the project. Hundreds of people were now working on a project of unprecedented complexity: [9000 superconducting magnets](#) to be produced, the [world's largest cryogenics system](#), an ultra-sophisticated power-supply system, and millions of sensors and other components manufactured by around a hundred companies worldwide. Production in industry proved to be particularly arduous, with inevitable technical and financial ups and downs.

On 10 September 2008, [the LHC started up](#) in front of the world's cameras. Unfortunately, nine days

later, a faulty interconnection brought the machine to a halt, causing major damage over 600 metres. After 14 months of repairs and improvements, the LHC was [successfully restarted](#) and the [LHC research programme began on 30 March 2010](#). Since then, the machine has gained in energy and luminosity, enabling many crucial advances in our understanding of the infinitely small.

Recollections

Lyn Evans came to CERN in 1969 as a research fellow, before becoming a staff member in 1971. He was LHC Project Leader from 1994.

“I’d never wanted a public switch-on for the LHC, but there’d been a big build-up and there was this nonsense about [blowing up the Universe](#). It was at a time when the web and social media were sufficiently well established for news to travel widely, and there was enormous concern. So, we had no choice, we couldn’t just get beam and then announce it. But there were risks to being so public: the night before the switch-on, the cryogenic system stopped working and we only recovered it in the morning of 10 September 2008, just before we could start up the beam!

Nobody could have imagined it would work so quickly. For [LEP](#), in 1989, it took two months to commission. For the LHC, it took just two hours!

The screen showed one spot when the beam entered the LHC and it was supposed to show a second spot when the beam made a full turn. I thought it hadn’t worked, because there was a delay, but then we saw the two spots, even three spots as it had gone around twice. The room erupted in applause, it felt like a football match!

The LHC project had spanned the mandates of four Director-Generals and each of them was perfectly timed. [Llewellyn Smith](#)’s diplomacy helped to get the project approved. [Maiani](#)’s physics background helped to convince a reluctant physics community to stop LEP, to make room for the LHC.

[Aymar](#)’s engineering background meant he fully understood the reality of the manufacturing issues, and he saw the project through to the end. When the physics programme began in 2010, [Heuer](#) was in charge. I had been asked to lead the LHC project in 1993, before it had even been approved. I guess I was naïve to have accepted, but I’d been division leader already and, alongside other tasks, I’d worked on initial designs since the 1980s, when the [first ideas for the LHC](#) came up after a meeting in Lausanne.

It was a 15-year construction project, with plenty of challenges along the way. Four suppliers went bankrupt. We had warehouses all over the Pays de Gex with components for magnet building, to have a buffer in our supply chain. The project was based on a just-in-time installation, but a one-year delay in the cryogenic line meant that we had [hundreds](#)

[of magnets above ground](#), waiting to be installed, in car parks, exposed to the elements. Every time I took an aeroplane, I’d look out of the window and see a sea of blue dipoles!



Lyn Evans (centre) stands next to CERN Director-General, [Robert Aymar](#) (right) as they watch the screens in the CERN Control Centre on 10 September 2008, the day of the first beam in the LHC. (Image: CERN)

Before the switch-on, we had taken seven sectors to full energy, but there hadn’t been time to do the eighth. There were 10 000 high-current joints between the magnets, and we knew that the weak point could be the interconnections. We’d even calculated the risk of failure and had a probability of 10^{-4} . But bear in mind that $10^{-4} \times 10\,000 = 1$ and, indeed, on 19 September 2008, nine days after the switch-on, we saw where the failure was. It happened around lunchtime; I was with Human Resources discussing contracts and got a call. I immediately went to the CERN Control Centre (CCC) and saw all the screens were red! One joint had failed and the relief valves weren’t big enough for the overpressure. It knocked about 50 magnets out of alignment. We had spares to make the repairs, but it took about a year.

Looking back now, I’d say that the incident was a blessing for the LHC experiments. It gave them extra time to use cosmic rays to calibrate their detectors and make simulations, so when we switched the LHC back on in 2010, we were able to ramp up much faster.

What the incident showed me is that there’s something about the people at CERN: when the going gets tough, the tough get going. Everyone sprang into action and used ingenuity to make it work. It’s like that at CERN, we quickly solve issues as they arise.”

Professor Costas Fountas elected as next President of the CERN Council

The CERN Council has announced that Professor Costas Fountas will become its 25th President beginning on 1 January 2025



*Professor Costas Fountas 25th President of the CERN Council.
(Image: CERN)*

Last week (week 39), the CERN Council elected Professor Costas Fountas as its 25th President, for a period of one year, renewable twice, with a mandate starting on 1 January 2025. He will take over from Professor Eliezer Rabinovici, who concludes his term at the end of December.

“Professor Fountas is an experimentalist of the highest calibre, who has played key roles in the CMS collaboration at the LHC and also in the former ZEUS collaboration at DESY in Germany. As Vice President of the CERN Council since 2022, his professionalism and advice have been of great benefit to discussions, and I am confident that he will make an excellent President at this very important time for the long-term future of the Organization” said Professor Eliezer Rabinovici, current President of Council.

Professor Fountas is Professor of physics and heads the high-energy physics group at the University of Ioannina in Greece. After completing his PhD at Columbia University in 1989, studying neutrino production at the Tevatron, he worked

on the data-acquisition electronics for Fermilab’s CCFR experiment. He then moved to the University of Wisconsin and worked on the trigger system of the ZEUS experiment at DESY. In 2000, he started work at Imperial College London and joined the CMS collaboration where, among several roles, he took responsibility for the design and implementation of the Global Calorimeter Trigger; his group also developed the first model of the CMS track trigger. Later, based at the University of Ioannina, he developed and built the CMS Barrel Muon Track Finder using cutting-edge FPGAs and optical-link technologies. At the LHC his physics interests have focused on jet cross sections and searches for new fundamental interactions.

Alongside his expertise in delivering detector systems and conducting physics analyses, Professor Fountas has held several management roles in CMS and been a member of many councils and advisory committees. He was appointed Greek scientific delegate to the CERN Council in 2016 and Vice President of Council in 2022.

“My focus will be to support the CERN Management and the experiments so as to ensure that the High-Luminosity LHC is completed successfully and in a timely manner. I will also make sure that discussions on the next major project at CERN are held in such a way that everybody has a voice. It is a critical time for CERN, and as President of Council, my commitment will be to do everything I can to bring consensus and guarantee the brightest future possible for the Organization,” said Professor Costas Fountas.

CERN Council is the supreme decision-making authority of the Organization, composed of delegates of all its twenty-four Member States.

NA62 experiment at CERN observes ultra-rare particle decay

In the Standard Model of particle physics, the odds of this decay occurring are less than one in 10 billion



The NA62 experiment at CERN. (Image: CERN)

Geneva, 25 September 2024. At a [seminar](#) held at CERN this week, the [NA62](#) collaboration reported the unequivocal confirmation of the ultra-rare decay of a positively charged kaon into a positively charged pion and a neutrino–antineutrino pair. Experiments including NA62 have previously measured and [seen evidence](#) of this process, but this is the first time it has been measured with a statistical significance of [five standard deviations](#), crossing the threshold traditionally required to claim a discovery in particle physics.

Denoted by $K^+ \rightarrow \pi^+ \nu \bar{\nu}$, this decay is among the rarest particle processes ever observed: in the [Standard Model](#) of particle physics, less than one in 10 billion positively charged kaons are predicted to decay in this way.

“This observation is the culmination of a project that started more than a decade ago,” says NA62 spokesperson Giuseppe Ruggiero. “Looking for effects in nature that have probabilities of happening of the order of 10^{-11} is both fascinating and challenging. After rigorous and painstaking work, we have finally seen the process NA62 was designed and built to observe.”

But why are physicists looking for a process that occurs so rarely? The reason is that theoretical models suggest that the $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ decay is extremely sensitive to deviations from the Standard Model prediction, making it one of the most interesting processes to search for evidence of new physics beyond the Standard Model.

Analysing data collected by the NA62 detector between 2016 and 2022, the NA62 researchers measured the fraction of K^+ that decay in this way to be $13.0_{+3.3}^{-2.9} \times 10^{-11}$. With a relative precision of 25%, this is the most precise measurement of the $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ decay to date.

The result is about 50% larger than the Standard Model prediction but is compatible with it given the overall uncertainty. With data taking ongoing, NA62 is set to be able to test the possibility of new physics in this decay within the next few years.

“Searching for hints of new physics in this decay requires more data, but this result is a leap forward and further strengthens the strong interest in this line of research,” says NA62 physics coordinator Karim Massri.

In the NA62 experiment, kaons are produced by slamming a high-intensity proton beam from CERN’s [Super Proton Synchrotron](#) into a stationary target. As a result, almost a billion secondary particles are produced each second, and these fly into the NA62 detector. Of these particles, about 6% are positively charged kaons. NA62 precisely detects the decay products of kaons, identifying and measuring all the particles produced except the neutrinos, whose presence is deduced from their missing energy.

Crucial to this result was the data from 2021 and 2022, which was taken following the completion of detector upgrades that enabled NA62 to operate at 30% higher beam intensities. Combined with improvements to data-analysis techniques, these hardware upgrades enabled the collection of signal candidates 50% faster than before, while adding new tools to suppress the background processes that could mimic the $K^+ \rightarrow \pi^+ \nu \bar{\nu}$ decay.

“This measurement relies on identifying the one-in-10-billion K^+ decay that is our signal and making sure it is not one of the other 9 999 999 999 decays that can mimic the signal,” says lead data analyst Joel Swallow. “The whole NA62 collaboration has made this almost impossible result possible.”

ALICE probes the strong interaction three-body problem

With new measurements of hadron–deuteron correlations, the ALICE collaboration explores the strong interaction of three-body systems at the LHC

In an article recently published in [Physical Review X](#), the ALICE collaboration presented its studies of correlations in the kaon–deuteron^[1] and proton–deuteron systems, opening the door to precise studies of the forces in three-body nuclear systems.

A fundamental force is typically described as an interaction between two objects. Extending this to more complicated systems is not always trivial. The description of strongly interacting three-hadron systems is key to understanding many phenomena in modern nuclear physics, such as the structure of nuclei, properties of high-density nuclear matter and the composition of neutron star cores.

Proton–proton collisions at the LHC produce a large number of particles that are emitted very close to each other, at distances of about 10^{-15} m (a femtometre). It is interesting to explore whether they influence each other in any way before spraying off in all directions. If two particles are produced close to each other and with similar momenta and direction, the pair can be subject to quantum statistics, Coulomb force and strong interaction. If one of the pair is a deuteron, then a system with a deuteron and another hadron, like a proton or a kaon, is effectively a three-body system. Thus, the measurement of correlations between deuterons and kaons or protons is expected to reveal the interactions of three-body systems.

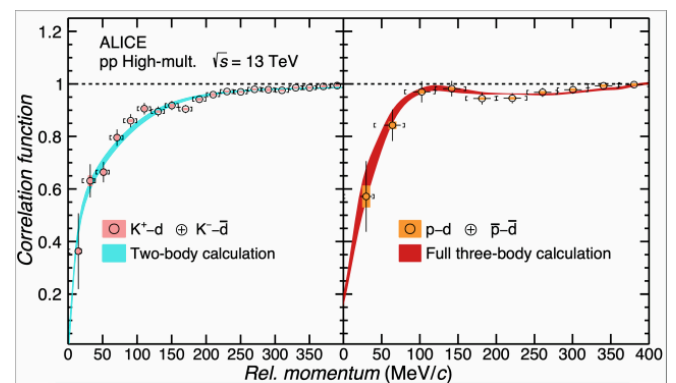
The ALICE collaboration utilises its excellent particle identification capabilities to study these correlations in high-multiplicity proton–proton collisions at a centre-of-mass energy of 13 TeV. The result is a correlation function that measures how the probability of finding two particles with certain relative momenta differs from what would be expected if their momenta were completely independent or uncorrelated. In the absence of correlation, the value of the function is unity. A value above one indicates attractive interaction, whereas a value below one indicates repulsive interaction.

The correlation functions for both the kaon–deuteron and proton–deuteron systems are below

unity for low relative transverse momenta (as seen in the figure below), indicating an overall repulsive interaction. The analysis of the kaon–deuteron correlation shows that the relative distances at which deuterons and protons or kaons are produced are quite small, around 2 fm.

The kaon–deuteron correlations are well described with an effective two-body model that incorporates both the Coulomb interaction and strong interaction between the kaon and the deuteron. In contrast, the same effective two-body approach fails to describe the proton–deuteron correlations, necessitating a full three-body calculation that accounts for the structure of the deuteron. An excellent data description is achieved using theoretical calculations that account for both two- and three-body strong interactions. This demonstrates the sensitivity of the correlation function to the short-range dynamics of the three-nucleon system.

The correlation measurements at short distances constitute an innovative method to study three-body systems at the LHC, with the potential to extend such studies to other hadrons. It is envisaged to apply a similar approach to data from LHC Runs 3 and 4 to investigate three-baryon systems in the strange and charm sectors, which are otherwise experimentally inaccessible.



Measured correlation function as a function of relative momentum in (left) kaon–deuteron fitted with effective two-body calculation and (right) proton–deuteron fitted with full three-body calculation. (Image: ALICE/CERN)

^[1] Deuteron is the atomic nucleus of deuterium, composed of a neutron and a proton held together by the strong interaction.

The new CERN Library, one year on

Improved accessibility, events, audiobooks and more – find out what the new CERN Library has to offer you



The bookshop of the CERN Library. (Image: CERN)

One year ago, after 12 months of extensive renovation work, [the new CERN Library was inaugurated](#). Discover what the new Library has to offer, as well as the many services that make reading and working easier. Psst: to celebrate our first year, visit the CERN Library to pick up a free tote bag, stickers and bookmarks.

1. Accessibility

Over the past year, the Library has stepped up efforts to improve the accessibility of its services and to make reading and working more comfortable. For example, you now have access to magnifying glasses, noise-cancelling headphones and earplugs, plus a Kuppel chair for better sound isolation. Find out more about [accessibility at the Library](#).

2. New features in the [Library catalogue](#)

Librarians and colleagues from the IT department have also developed new tools, such as a self-checkout option to borrow books, and they have integrated the Library catalogue in MapCERN to help users easily locate books on the Library shelves. You can now also find audiobooks and videos in the catalogue.

3. Events

In the renovated space, the Library hosts two new series of events: Meet the Author, which has already welcomed authors Nicole Yunger Halpern and Chris Quigg, and The Voice of Science, short talks related to science communication, with

invitees such as Zach Weinersmith and AstroKatie. Check out our [upcoming events](#).

4. Workplace

The [Library \(Building 52, first floor\)](#) is open 24/7 and is a quiet place to study, equipped with modern and comfortable furniture, 40 workstations (of which four have extra screens), power plugs, a private sound-proof cabin for phone/video calls, lockers and a water fountain. Additionally, the Library staff are available to assist you, from Monday to Friday, from 9 a.m. to 6 p.m. You'll find them at the Welcome Desk.

5. Collections

Over the past year, more than 17 200 new resources have been added to the Library catalogue. Depending on your needs and preferences, you can choose from books, e-books, journals, audiobooks or videos. In total, more than 16 500 physical books are available in the Library and more than 40 500 are available on request from our closed stacks, while 121 000 e-books, more than 6800 videos and more than 2900 audiobooks are available or requestable online. Search our [collection](#) and [latest additions](#), as well as the [full list of resources](#).

6. Services

Can't visit the Library? No problem, you can request books from the [catalogue](#) to be delivered directly to your office. Need a book that is not available in the Library? No problem, we can order it for you via our interlibrary loan service. Need a book or standard that you want to keep in your service for regular use? No problem, we'll take care of the ordering process. Just fill in our [request form](#).

7. Bookshop

Did you know that the Library also has a bookshop? Stop by to get a copy of your favourite reference book or pick up a CERN book as a souvenir or gift. Browse our [bookshop collection](#) to discover our titles.

8. Content curation

Behind the scenes, CERN's Library team is working hard to ensure that CERN publications are accurately listed and described in repositories. Every year, around 12 to 18 000 articles, conference papers, photos, etc. are curated by our

devoted Scientific Information Service team on the [CERN Document Server \(CDS\)](#) and [INSPIRE](#).

To find out more about what the Library has to offer, visit [our website](#) and subscribe to our [newsletter](#).

CERN Library

2024 Beamline for Schools winners at CERN and DESY

CERN and DESY welcomed the winners of the 2024 Beamline for Schools competition from 12 to 26 September



“Sakura Particles”, the team comprising pupils from different high schools in Japan, and “Mavericks”, the team made up of pupils from two schools in Estonia, were welcomed on stage during the [CERN70 Community Event](#). (Image: CERN)

From 12 to 26 September, CERN and [DESY](#) (the German electron synchrotron facility) welcomed the [winners](#) of the 2024 [Beamline for Schools](#) (BL4S) competition. Students from Estonia, Japan and the USA carried out their own [experiments](#) using accelerator beams at the facilities. “Mavericks”, the team from Estonia, and “Sakura Particles”, the team from Japan, came to CERN, while “SPEEDers”, the team from the United States, was invited to DESY.



“SPEEDers”, the team formed of pupils from a high school in the United States, carried out its experiment at a DESY beamline. (Image: DESY/BL4S)

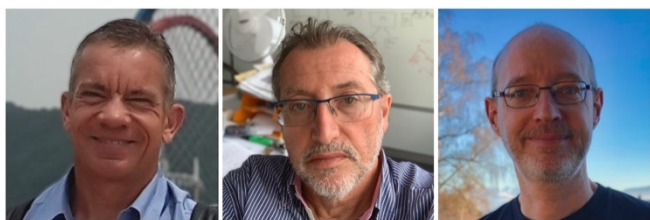
BL4S started in 2014 in the context of CERN’s 60th anniversary. Over the past ten years, more than 20 000 pupils from all over the world have taken part in the competition and 25 teams have been selected as winners. The participation rate has been rising consistently over the years, with a record 461 teams from 78 countries submitting an experiment proposal in 2024.

The fruitful collaboration between CERN and DESY started in 2019; this year marks its fifth anniversary.

Beamline for Schools is an education and outreach project funded through the [CERN & Society Foundation](#) by individual donors, foundations and companies.

Anaïs Schaeffer

Three new members elected to the Senior Staff Advisory Committee (“the Nine”) in 2024



Klaus Hanke (left), Ludovico Pontecorvo (centre) and Thorsten Wengler (right), the new members of the Senior Staff Advisory Committee. (Image: CERN)

The electronic voting process for the [Senior Staff Advisory Committee](#) (known as “the Nine”) closed at midnight on Tuesday, 10 September 2024. This time a second round of voting was held, as two of the candidates received the same amount of votes during the first round. The Senior Staff Advisory Committee was created in 1981 to serve as a channel of communication between the senior staff (grade 8 and above) and the Director-General. It is made up of nine members elected by the senior staff for a period of three years. The Nine share the ideas and feedback of the senior staff with the Director-General and offer advice on questions concerning scientific activities, organisational matters and use of resources. Elections for the Nine are held every year, ensuring an annual rotation of members.

In August and September 2024, out of the 561 senior staff members eligible to vote, 310 voted in the first round and 186 in the second.

Candidates stood for election for Electoral Group 1 (research physicists and applied physicists in the EP or TH departments) and for Electoral Group 2 (members of IT, RCS, SCE, HSE, ATS, EP and TH, excluding those with the following benchmark jobs: applied physicist, principal applied physicist, research physicist, theoretical physicist, principal research physicist or principal theoretical physicist).

The mandates of Klaus Hanke, Ludovico Pontecorvo and Thorsten Wengler will run from

September 2024 to August 2027, replacing the outgoing members David Barney, Christophe Delamare and Christoph Rembser.

Markus Brugger has been appointed as the new spokesperson for one year, starting in September 2024. The Nine now consists of the three newly elected members together with [end of mandate in brackets]:

- Sophie Baron [2026]
- Giovanna Vandoni [2026]
- Marzia Bernardini [2025]
- Markus Brugger [2025]
- Cécile Curdy [2025]
- Niko Neufeld [2025]

We wish to congratulate the newly elected members and warmly thank all the other candidates. Special thanks also go to our polling officer, Alberto Pace.

Christoph Rembser and Markus Brugger, outgoing and incoming spokespersons of the Nine

The Nine seeks input for topics to be investigated throughout the year, so we encourage you to get in touch, either by sending an email to <mailto:the-nine@cern.ch> or by contacting one of our members. We meet every Thursday from 12.15 to 1.45 p.m. In 2025, our weekly meetings will move to Tuesdays (same time slot). You are welcome to attend the first 15–20 minutes of our meeting to present your topic in person. Details of previous topics are available on [the Nine webpage](#). Furthermore, we are aiming to organise monthly visits to different CERN departments in order to engage in discussions with senior staff; we therefore invite all interested senior staff colleagues to contact one of us, either in person or by phone/email, for more information or to schedule a visit.

Computer Security: Happy Birthday, Virus!

35 years ago, on 9 October 1989, the CERN Computer Security team issued for the first time a “Computer Virus Warning to all users of IBM Compatible PCs” (see the memorandum on the right). While it was not even the first virus the world had seen (“Creeper” of 1971 for DEC mainframes inter-connected via the ARPANET, “Elk Cloner” of 1982 targeting Apple IIs via infected floppy disks, and [“©Brain” going after DOS-based PCs](#) were doing their evil deeds earlier), this 1989 virus was a precursor without a name for the ransomware attacks of today. And it warranted a CERN-wide warning. Only decades later did more powerful viruses, i.e. “Slammer” and “Blaster” of 2003 and 2005, respectively, hit harder. Things work the same way today, 35 years later. A virus, when activated, crawls through your documents and data, threatens to erase all the files from the hard disk and asks for “ransom” money (or threatens to delete that data if the payment is not paid – [if it is even enough](#)). The only innovation compared to the past is that data nowadays is also encrypted to better make the point. [Newer variants even exfiltrate that data and threaten to publish it](#). Your private photos and personal documents, all being made public. To shame you. To pressure you. To make you pay. And there is another variation compared to the past: the attack vector. While it used to be infected floppy disks that were slowly passed on from one PC to another, attacks today require either a lack of vigilance on the part of users who click on random URLs, links, [QR codes](#) or similar and, subsequently, compromise their computer, or so-called zero-day vulnerabilities in the operating system, browser, mail client or any other application digesting publicly shared documentation.

C037
MEMORANDUM
To: all subscribers of the MIS Bulletin
From: MIS/OCS 89.10.09
Subject: Computer virus (Texte français au verso)
VAN ELSLANDE Jean-Louis
PE

Computer Virus Warning to all users of IBM Compatible PCs

What we know:

The press have announced the existence of a virus attacking IBM PC compatibles (IBM, Olivetti, Toshiba, etc. etc.) running MS-DOS and maybe other operating systems.

The virus is supposed to act by looking at the computer clock/calendar and to erase all files from the hard disc if the date is October 12 or later. The virus has spread from The Netherlands, and comes as a set of three routines, called Datacrime 1, 2 and 3.

We do not know whether CERN machines have contracted this virus or not, we do not know how serious the threat is, nor whether antidotes that have been announced really work.

What to do:

Do not panic. October 12 is only this Thursday, so you have some time. Follow these steps:

- Before Thursday, identify on your hard disc all documents and data files that are important to you. Do not attempt to back up all files, this would take too much effort. Do not back up system or application programs such as Windows, Excel, Word etc, since you can always reinstall these. Concentrate on the data, text and graphics that you created.
- make copies onto disquettes of these files. Write-protect the disquettes.
- make a duplicate copy of these files. Write-protect the disquettes.
- keep one set of disquettes in a locked-up place, preferably in a room under fire protection.
- keep the other set in a different place.
- Continue to work normally, but every evening make a copy of the documents you changed that day.
- Do not use the backup copies you have made, especially if you need them urgently, before having contacted us.
- On and after Thursday, if you think your computer behaves strangely, switch it off immediately, and do not restart it. Then contact either M. Sheehan (8138) or B. Henningsen (7163).

We have some virus detection programs and are in contact with the Dutch police to obtain from them the special program that eradicates this specific virus.

For further information: please do not flood us with calls to hear more information, but watch the general and the MIS news on the central VM system (if you do not have a means of connecting to VM, there is almost certainly someone in your building who can do so, inform yourself there).

Please also observe that the steps outlined above are what we have recommended in the past that you do normally anyway as part of your habits of using a personal computer.

Cross your fingers!

R. Cailliau
DD/OCS

If one of the criteria is fulfilled, the malicious virus does its malicious deed. On the defence side, the baseline, however, is also still the same. “Identify [...] all documents and data files that are important to you”; “Make copies”; “We have some virus detection programs”; “Cross your fingers” – not much has changed compared to today. And we happily repeat these basics for protecting your computer, laptop or smartphone here again:

- Always keep your operating system up to date. Give it some time (during the night) to patch. At least this way you protect it against any known vulnerability;
- Ditto for any other application, in particular your mail client, office application and browsers;
- Run a decent anti-malware software like the one that [CERN provides for Windows and Mac operating systems](#);

- STOP – THINK – DON'T CLICK in order to avoid any drive-by infection when presented with a malicious URL, link, [QR code](#), attachment, etc.;
- Have a decent [back-up of all your crown jewels](#), i.e. all photos, documents and files that are valuable to you. In fact, CERN-related, professional data and documents should be stored centrally in CERN's IT services (CERNBox,

EDMS, GitLab, Indico, etc.) and not reside solely on your computer. For your personal stuff, read this bullet point again, and go buy an external hard disk!

- Cross your fingers.

Computer Security Office

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

1–6 Oct | Knowledge Sharing | Mont Blanc bridge | CERN70 | EN & FR

[CERN flags are flying on the Mont Blanc bridge in Geneva](#)

1–31 Oct | Knowledge Sharing | Rotonde du Mont-Blanc | CERN70 | EN & FR

[CERN exhibition in the Rotonde du Mont-Blanc in Geneva](#)

10 Oct 18:00 | Knowledge Sharing | Online | CERN Alumni | EN

[News from the Lab | CERN openlab](#)

11 Oct 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN

[L&D Micro-talk 12 – Demystifying creativity](#)

12 Oct 08:00 | Knowledge Sharing | Idea Square | CERN Micro Club and CERN Women in Technology | EN & FR

[Python for Business – Reskilling training for women](#)

14–18 Oct | Knowledge Sharing | Council Chamber | Academic Training Lectures | EN

[Signals, Noise and Signal processing in Particle Detectors](#)

15 Oct 17:30 | Knowledge Sharing | Council Chamber | Staff Association | FR

[« Qui est légitime à parler du temps », par Etienne Klein](#)

17 Oct 12:00 | At CERN | Filtration Plant (222/R-001) & online | CERN Medical Service and partners | EN & FR

[Stroke Awareness Campaign: talks by HUG experts & young stroke survivors / Campagne de sensibilisation](#)

[aux AVC: présentations par les experts HUG et témoignage de jeunes survivants d'un AVC](#)

17 Oct 13:30 | Knowledge Sharing | CERN Library | Archives, Library and Open Science events | EN

[Meet the author of "Relic Gravitons"](#)

18 Oct 14:00 | Knowledge Sharing | Main Auditorium | Commemorations | EN

[Symposium to celebrate Carlo Rubbia's 90th birthday](#)

18 Oct 16:00 | Knowledge Sharing | BE Auditorium | BE Seminars | EN

[PUMA Experiments](#)

23 Oct 14:00 | Physics | Council Chamber | Theory Colloquia | EN

[The Higgs Boson: From Theory to Experiment](#)

25 Oct 14:30 | Knowledge Sharing | Online | CERN Alumni | EN

[Moving Out of Academia to Aviation & Aerospace](#)

31 Oct 19:00 | Knowledge Sharing | CERN Science Gateway | Public event | FR (simultaneous interpreting in EN)

[Dark Matter Day 2024](#)

Registration now open

8 Nov 11:00 | Knowledge Sharing | The Globe of Science and Innovation | IdeaSquare | EN

[IdeaSquare 10th anniversary](#)

27 Nov–6 Dec | Experiments | CERN | DRD1 | EN

[DRD1 Gaseous Detectors School](#)

This is a curated list of events relevant to the CERN community. More events are available here:

Feasibility Study for a possible Future Circular Collider (FCC) continues on the ground

As part of the Feasibility Study for the Future Circular Collider (FCC) project, geological field studies are now under way to determine the nature of the subsoil and to better understand the stability of the soil

The aim of the [FCC Feasibility Study](#), which is scheduled to be completed in 2025, is to determine the technical and financial viability of the [FCC](#) at CERN, with a particular focus on its geological aspects, the environmental impact, the infrastructure design, the civil engineering and the detectors and the R&D for the technologies

needed to ensure the efficiency and durability of the proposed colliders.

With this in mind, in 2023 CERN conducted [initial assessments on the ground](#) in order to refine the existing geological and seismic data, as well as data regarding flora and fauna for the purpose of conservation. This first phase involved a visual inspection of the areas of land concerned.

to determine at exactly what depth it could be installed.

The results of these analyses will make it possible to refine the placement scenarios and determine which ones should be prioritised if the project is approved, taking into account both the environmental objectives for the areas on the surface and the underground constraints.

Based on the results of the Study, CERN's Member States will make a decision about the project as a whole and, in particular, about the prospects for commissioning the FCC in the 2040s.



The above document was sent out to the inhabitants of the communes concerned by the studies under way to explain the nature of the studies and the methods that are being employed. For more information, visit the [FCC Feasibility Study website](#).

In the second phase of the Study, which has just started, the focus is on the stability of the subsoil. This is a crucial step, as the project includes plans to bury the collider approximately 200 metres underground. Before going any further with the analyses, it is vital to check the nature of the soils along the entirety of the planned route to see if they are suitable for this kind of infrastructure and

For further information about the investigations under way in the canton of Geneva (Switzerland) and in the departments of Ain and Haute-Savoie (France), and about the related potential noise pollution (linked, in particular, to the use of lorry-mounted seismic vibrators), visit the [“CERN and its neighbours”](#) website and the [FCC Feasibility Study website](#) (in French).

Take action to prevent strokes

Participate in CERN's stroke awareness campaign on 17 October

STROKE AWARENESS CAMPAIGN
Campagne de sensibilisation à l'AVC

17 October 2024
17 octobre 2024

10 a.m. to 3 p.m.:
Interactive information stands - various locations
R1/Main Building
Stands d'informations interactifs - lieux divers
R1/Bâtiment principal

12 p.m. to 1.30 p.m.
Talks by HUG experts & young survivors' testimonials (filtration plant 222/R-001)
Présentations par les experts des HUG et témoignage de jeunes survivants d'un AVC

For more information/ plus d'informations sur :
hse.cern/stroke

CERN HSE Occupational Health & Safety and Environmental Protection unit

novae HUG FRAGILE GENEVE World Stroke Organization

When it comes to medical emergencies, few conditions strike as suddenly and as devastatingly as a stroke. Strokes occur when the blood supply to your brain is interrupted and they are life-threatening emergencies. If not treated quickly, there may be significant after-effects. The best known is hemiplegia (paralysis of one side of the body), but there are many others, such as speech disorders, sensory disorders, emotional disturbances and more. According to the [World Stroke Organization](#), there are 12.2 million new strokes globally each year. What's more, strokes concern us all, with one in four people over the age of 25 predicted to suffer a stroke during their lifetime.

[World Stroke Day](#) is observed every year in October to raise awareness of the serious nature and high rates of strokes and of how to prevent and treat them, and to ensure better care and support for survivors. CERN's Medical Service has teamed up with

several local partners for an event on Thursday, 17 October, featuring interactive information stands, in order to help us all to better understand and prevent strokes.

The [Hôpitaux Universitaires de Genève](#) (HUG) will be present to explain the warning signs of a stroke – using virtual reality goggles to create a tangible experience – and how strokes are managed in the emergency room.

The Medical Service team will offer you tips and tricks on mental health awareness and how to reduce stress, a key factor in the occurrence of strokes, such as how to use the cardiac coherence technique.

Medical staff will be on hand to measure your blood pressure, an important health indicator in stroke prevention.

[Novae](#) will host a stand to explain the impact of nutrition on our health and offer tips to better prevent strokes, while [FRAGILE](#) and the [World Stroke Organization](#) will cover specific information on the support and resources available through their networks.

Finally, the [CERN Knowledge Transfer group](#) will demonstrate two ongoing projects linked to strokes, namely the [CAFEIN](#) platform and the [TRUSTroke](#) project.

The campaign programme includes a fascinating [lunchtime conference](#), with speakers from HUG discussing recent advances in stroke medicine. It will also feature testimonials from two young stroke survivors, [Elise Mathy and Louis Gustin](#), who will share their personal experience in a talk entitled "Journey into aphasia".

Visit the stands in Restaurant No. 1 and the foyer of Building 60 from 10 a.m. to 3 p.m. and don't miss the conference taking place from 12 noon to 1.30 p.m. in the filtration plant, Building 222. Sign up to attend the conference at this [link](#).

For more information about strokes and the full campaign programme, go to: <https://hse.cern/stroke>.

HSE unit

Switzerland: adoption of the amended LERI (*Loi sur l'encouragement de la recherche et de l'innovation*) and the *Plan sectoriel fédéral*

On Friday, 27 September 2024, the Swiss Parliament adopted [amendments to the *Loi sur l'encouragement de la recherche et de l'innovation*](#) (Federal Act on the Promotion of Research and Innovation (LERI)). In accordance with the wishes of the Swiss Federal Council, a legal basis now exists for the development of a *Plan sectoriel fédéral* (Federal Sectoral Plan) regarding CERN's buildings and installations.

Under this plan, which is currently being developed, the Federal Council intends to clarify the administrative procedures concerning land-use planning and improve the scheduling certainty of strategically important CERN projects that entail territorial development.

CERN participated in the consultation opened in March 2023 with a view to amending the LERI and

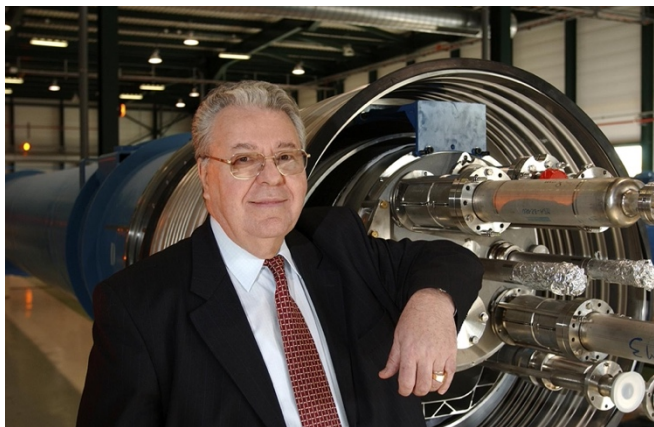
subsequently appeared before the Science, Education and Culture Committee of the National Council (Lower Chamber of Parliament) in May 2024.

The adoption of the amended LERI by the Swiss Parliament is an important milestone. CERN appreciates the commitment of the Swiss authorities, in their role as Host State, to science, as is reflected in the Federal Council's [Foreign Policy Strategy 2024–2027](#).

CERN will continue to cooperate closely with the authorities in the pursuit of their work with the aim of improving support for the Organization's future projects, including the Future Circular Collider (FCC), in the event of its realisation, in accordance with the [Federal Council's decision of 10 December 2021](#).

Obituaries

Robert Aymar (1936 – 2024)



Robert Aymar, former CERN Director General, in front of a LHC dipole magnet. (Image: CERN)

Robert Aymar, CERN Director-General from January 2004 to December 2008, passed away on 23 September at the age of 88. His term of office was marked by the completion of construction and the first commissioning of the [Large Hadron Collider](#) (LHC). His experience of complex industrial projects proved to be crucial, as the CERN teams had to overcome numerous challenges linked to the LHC's innovative technologies and their industrial production.

A specialist in plasma physics and its application in thermonuclear fusion, Robert Aymar had previously led major scientific projects based on innovative technologies. He directed the Tore Supra tokamak fusion research programme from 1977 to 1988, and then, from 1994, the International Thermonuclear Experimental Reactor ([ITER](#)) project, based on technologies developed for Tore Supra.

As Head of the Material Sciences division at the Commissariat à l'énergie atomique et aux énergies

alternatives (French Alternative Energies and Atomic Energy Commission, CEA) from 1990 to 1994, he set out to bring together the physics of the infinitely large and the infinitely small, as well as the associated instrumentation, in an institute that has now become the Institut de recherche sur les lois fondamentales de l'Univers (Institute for Research into the Fundamental Laws of the Universe, [CEA IRFU](#)). He served on many national and international committees, and in 1993 he chaired the LHC External Review Committee, whose recommendation proved decisive in the [project's approval](#). In 2001, the CERN Council called on his expertise once again by entrusting him with the chairship of the [LHC programme's External Review Committee](#).

When he took over as Director-General of CERN in 2004, the construction of the LHC was well under way, but there were many industrial and financial challenges still to overcome. During his tenure, the machine was completed and the [first beams circulated](#). That first start-up in 2008 was followed by a major technical problem that led to a shutdown lasting several months. But the LHC had demonstrated that it could run, and in 2009 the machine [was successfully restarted](#).

Robert Aymar's term of office also saw a simplification of CERN's structure and procedures, aimed at making the Laboratory more efficient. He also set about reducing costs and secured additional funding to complete the construction and optimise the operation of the LHC.

A full obituary will appear in the [CERN Courier](#).